A self-contained formulation of Theorem C.2

Problem. *For $n \geq 0$, define*

$$M_n(\lambda) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mathrm{Cat}(r) \lambda^{n-r}, \qquad \mathrm{Cat}(r) = \frac{1}{r+1}\binom{2r}{r}.$$

Define $M_n^{(1)}(\lambda)$ by

$$\begin{aligned} M_n^{(1)}(\lambda) = \lambda & \sum_{\substack{a,b,c \geq 0 \\ a+b+c+3=n}} (a+1) M_a(\lambda) M_b(\lambda) M_c(\lambda) \\ & + \lambda \sum_{\substack{a,b,c,d \geq 0 \\ a+b+c+d+4=n}} (a+1) M_a(\lambda) M_b(\lambda) M_c(\lambda) M_d(\lambda). \end{aligned}$$

For $m \geq 0$, set

$$H_m(\lambda) = \big(M_{i+j+1}(\lambda)\big)_{0 \leq i,j \leq m}, \qquad H_m^{(1)}(\lambda) = \big(M_{i+j+1}^{(1)}(\lambda)\big)_{0 \leq i,j \leq m},$$

and

$$f_{0,m}(\lambda) = \det H_m(\lambda), \qquad f_{1,m}(\lambda) = \mathrm{Tr}\big(\mathrm{adj}(H_m(\lambda)) H_m^{(1)}(\lambda)\big).$$

Finally define $d_r(\lambda)$ by

$$d_0(\lambda) = 1, \qquad d_1(\lambda) = \lambda, \qquad d_{r+2}(\lambda) = \lambda d_{r+1}(\lambda) - \lambda d_r(\lambda),$$

and define $g_m(\lambda)$ by

$$d_{m+1}(\lambda) = \lambda^{\lceil (m+1)/2 \rceil} g_m(\lambda), \qquad g_m(0) \neq 0.$$

Theorem C.2. *For every $m \geq 0$,*

$$g_m(\lambda) \mid \big(f_{1,m}(\lambda) - f_{0,m}'(\lambda)\big) \qquad \textit{in } \mathbb{Q}[\lambda].$$